CrimeSphere An Adaptive Evidence Centered Agentic AI Framework for Murder Investigation

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***Abstract*—Murder investigations rely on information that arrives from many sources at different points in time, including FIR records, victim and suspect details, photographs, CCTV footage, forensic and post-mortem reports, and historical case records. Existing AI-assisted approaches, including retrieval-augmented generation and language-model-based assistance, generally treat a case as a fixed collection of documents to be searched, leaving the investigator to judge how newly added evidence changes the interpretation of the case. This paper presents CrimeSphere, a framework that maintains an adaptive case state for an ongoing murder investigation and updates this state as new evidence becomes available. CrimeSphere compares the current case state against expected categories of investigation information to flag potentially missing evidence, retrieves relevant historical cases and supporting documents, and uses an investigation agent to generate recommendations linked back to the evidence that supports them. The investigator reviews and validates every output; the system does not make determinations about guilt, punishment, or legal responsibility. We describe the proposed architecture and report a controlled evaluation against a static retrieval baseline across evidence-gap detection, retrieval precision, recommendation traceability, and recommendation relevance.**

***Index Terms*—murder investigation, adaptive case state, evidence completeness, agentic AI, retrieval-augmented generation, traceable recommendations, human-in-the-loop**

I. Introduction

A murder investigation is information-intensive from the moment it opens. An FIR is registered, and the case grows to include victim and suspect details, crime-scene photographs, CCTV footage, weapon information, forensic and post-mortem reports, and, over time, records of similar past cases. Unlike a document written once and finalized, this information keeps arriving, and different pieces become available at different points in the investigation. The evidence is also heterogeneous: structured FIR fields sit alongside scanned reports, images, and video, and the investigator has to interpret all of it together. As a case grows, tracking what has already been established and what remains outstanding becomes progressively harder to do manually, particularly in cases that stay open for weeks or months.

A range of AI-based systems have been proposed to support parts of this process, including database-driven case investigation and case-based reasoning [1], retrieval-augmented summarization applied to real homicide case files [2], language-model-augmented police assistants [3], graph-based representations of criminal case data [4], and knowledge-based reasoning over crime-reconstruction evidence [5]. These systems show that retrieval, summarization, and reasoning techniques can meaningfully assist an investigation, and each addresses a specific part of the workflow well. What most of them share is that they operate on the case as a static set of documents at the time of the query [2]. Retrieval-augmented generation itself was formulated around a language model paired with a fixed retrieval index [10], and follow-up work shows that even well-retrieved context is used unevenly depending on where it sits in a long prompt [14], which matters more once the case context is reconstructed repeatedly as new evidence arrives. When a system treats the case as fixed, the investigator is left to work out manually how a newly uploaded CCTV clip or forensic report should change the standing understanding of the case.

We propose CrimeSphere, a framework built around an adaptive case state rather than a static document index. As evidence is added, CrimeSphere updates the case state and re-evaluates it against categories of information typically expected in a murder investigation, retrieves historical cases and documents relevant to the current state, and uses an investigation agent to produce recommendations that reference the specific evidence supporting them. CrimeSphere is a

decision-support tool: it does not determine guilt or assign legal responsibility, and every output is reviewed by the investigating officer before any action is taken.

The contribution of this paper is not the use of an LLM, RAG, OCR, or similar-case retrieval individually, since each already appears in prior investigation-support and forensic-AI systems. The contribution is the combination of an adaptive case-state representation, evidence-completeness analysis over that state, and traceable agentic reasoning, applied to the murder-investigation workflow. We evaluate this design against a static-RAG baseline using a controlled set of 20 fictional investigation scenarios, and report results for evidence-gap detection, retrieval precision, recommendation traceability, and recommendation relevance.

II. Related Work

A. AI-Assisted Criminal Investigation. Ding [1] applies AI-based data processing to case databases for case-based reasoning and similar-case comparison. Skipanes et al. [2] evaluate summarization and memory-based RAG on two real homicide case files (roughly 3,700 documents, 13,000 pages), showing RAG scales to actual homicide data, though over an existing, unchanging document set. Kim et al. [3] introduce LAPIS, an LLM combined with a crime-investigation knowledge base for officer query answering. Bellandi et al. [4] propose graph embeddings for criminal investigation combining predictive precision with transparency. Wang et al. [5] propose a Bayesian-network model that generates investigative suggestions from case evidence. None of these systems is built around an explicitly evolving case state re-evaluated as new evidence is added during an active investigation.

B. Retrieval-Augmented Generation. Lewis et al. [10] combine a parametric language model with retrieval over a non-parametric index. Karpukhin et al. [11] propose dense passage retrieval, and Reimers and Gurevych [12] introduce Sentence-BERT embeddings suited to similar-case retrieval. Liu et al. [14] show language models use long contexts unevenly, relevant to how retrieved case material should be ordered. Es et al. [13] propose RAGAs for evaluating RAG systems without extensive human-labeled ground truth. CrimeSphere uses retrieval and RAG in the same general role as this line of work, but reconstructs context from an evolving case state rather than a fixed index.

C. Forensic and Digital Forensic AI. Ketsekioulafis et al. [6] and Tournois et al. [7] review AI across forensic disciplines, both flagging validation and interpretability as open concerns and noting current applications remain assistive rather than autonomous. Chernyshev et al. [8] and Wickramasekara et al. [9] survey LLMs in digital forensics, including hallucination risk and the need for expert review of model output before it is relied upon.

D. Language Models and Agentic Reasoning. Yao et al. [15] propose ReAct, interleaving reasoning with actions so a model can decide what to retrieve or invoke. Schick et al. [16] show a model can learn to call external tools without explicit supervision. Wei et al. [17] show intermediate reasoning steps improve multi-step performance; this is cited only as background on LLM reasoning behavior, and CrimeSphere does not expose or rely on any hidden chain-of-thought mechanism.

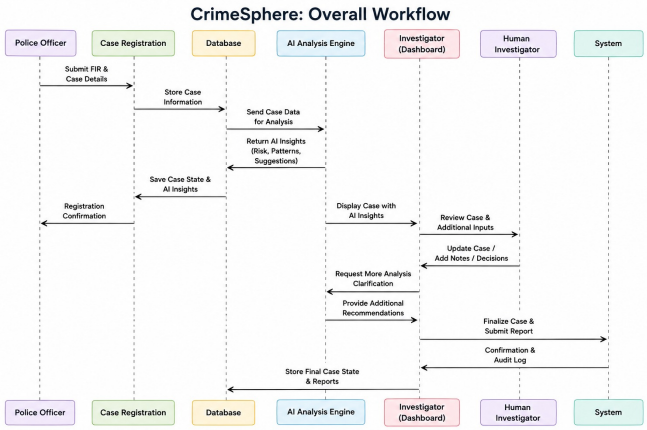


Fig. 1. Overall CrimeSphere investigation workflow.

E. Human-in-the-Loop and Explainability. Amershi et al. [18] give guidelines for human-AI interaction, including making capabilities and limitations clear and supporting efficient user correction. Ribeiro et al. [19] propose LIME for explaining individual model predictions. CrimeSphere's traceability mechanism (Section III-D) follows the same motivation: a recommendation is only useful to an investigator if its basis can be checked.

F. Research Gap. Existing work demonstrates progress across AI-assisted investigation, retrieval-augmented analysis, graph-based investigation, forensic AI, and LLM assistance [1]–[9], [15], [16], but does not fully address maintaining an explicitly evolving case state that coordinates evidence-completeness analysis, retrieval, agentic reasoning, and evidence-linked recommendations within one murder-investigation workflow. CrimeSphere is proposed to address this gap.

III. Proposed System

A. System Overview

CrimeSphere follows the workflow shown in Fig. 1: a verified police officer registers a murder FIR and case details, evidence is collected and uploaded as it becomes available, and this information is used to construct and continuously update a case state. The state is checked for evidence completeness, drives retrieval over historical cases and current information, and is passed to an investigation agent that produces recommendations linked to their supporting evidence. When new evidence is added, the case state updates and the cycle repeats.

Access is restricted to verified police officers. An officer supplies a police ID, registered phone number, and a police identity document. The document is processed with Tesseract OCR [20] to extract the printed police ID, which is compared against the entered ID and the police database record, together with the officer's authorized station. Verification success is followed by standard secure login through Spring Security and

TABLE I
CRIMESPHERE CASE INFORMATION CATEGORIES

Category	Information
FIR	FIR number, date, description
Victim / Suspect	Name, age, details, photograph, status
Crime	Location, date, time, weapon
Evidence	Images, CCTV, documents
Forensic	Forensic and post-mortem reports
Historical	Similar previous cases
External	Current news and public information

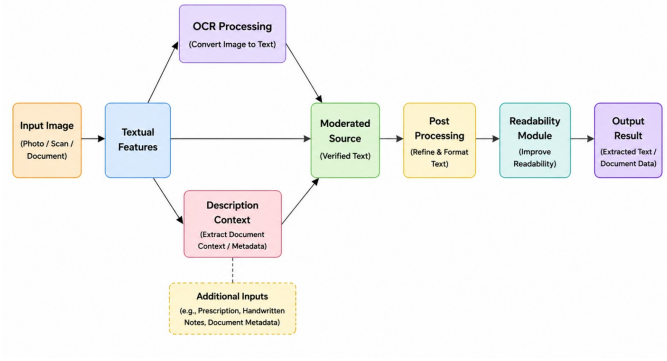


Fig. 2. Adaptive case-state evolution as new investigation information becomes available.

JSON Web Tokens. OCR and document verification are used here purely as supporting infrastructure and are not part of the research contribution.

A case is represented using the categories in Table I: FIR details, victim and suspect information, crime location and weapon information, evidence, forensic and post-mortem reports, historically similar cases, and relevant external information. This representation is populated incrementally as the investigation proceeds rather than required in full at FIR registration.

B. Evidence Management and Adaptive Case State

Evidence is uploaded and organized by type: photographs, CCTV footage, forensic reports, post-mortem reports, and general investigation documents. Each item is stored with metadata linking it to the case, which is what allows the traceability mechanism in Section III-D to reference a specific evidence item rather than the case file as a whole.

Unlike a static document index, the case state is explicitly updated whenever new evidence is added. Fig. 2 illustrates this: an initial state with FIR, victim, and suspect information is extended once CCTV footage is added, and again once a forensic report is added. Each full version of the state, not just the newly added item, is passed to the completeness check and the investigation agent, so reasoning takes the full current case context into account rather than treating new evidence in isolation.

C. Evidence Completeness Analysis

At each case-state update, CrimeSphere walks through the fixed category list in Table I and checks whether the current

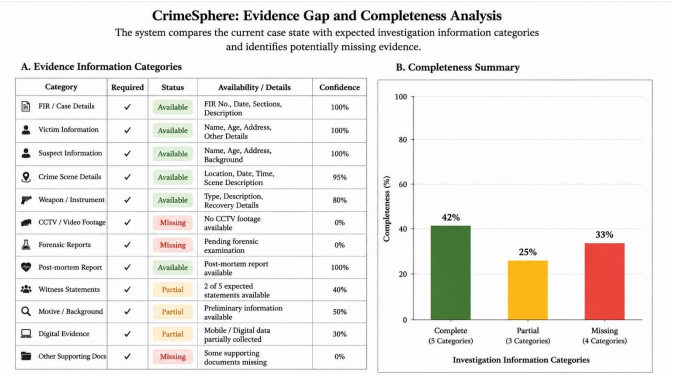


Fig. 3. Evidence completeness analysis for the current case state.

state contains an entry for each one. A category is marked available when at least one corresponding item has been entered, partial when only some of the expected sub-fields are present, and not-yet-entered when no matching item exists in the state. This distinction is deliberately weaker than a claim about the physical investigation: CrimeSphere only reasons over what has been recorded in the system, so a category marked not-yet-entered means the information has not been logged, not that the underlying evidence does not exist or was never collected.

The output of the check, illustrated in Fig. 3, is presented to the investigator as a status view across the expected categories rather than as a conclusion about the case. Categories flagged as not-yet-entered are intended to prompt the investigator to consider whether that information should be pursued next; the investigator decides whether and how to act on each flag, and CrimeSphere does not escalate or act on a flagged category on its own.

D. Traceable Recommendations and Human Validation

When the investigation agent produces a recommendation, the system attaches the supporting evidence, document, or record used to generate it, with a pointer to the relevant section where available. Fig. 4 shows an example: a recommendation to review the relationship between a suspect and a piece of evidence is linked to the specific evidence record it draws on, letting the investigator check the basis of a recommendation rather than accept it without verification. This follows the same motivation as prior work on explaining individual model outputs [19] and on keeping AI reasoning inspectable to the user [4], [18].

Every output produced by CrimeSphere, whether a completeness flag, a similar-case match, or an agent recommendation, is reviewed by the investigator before being acted on. A recommendation produced after a case-state update is linked to the evidence current at that point, so an earlier recommendation is not carried forward unchanged once the underlying case state has moved on.

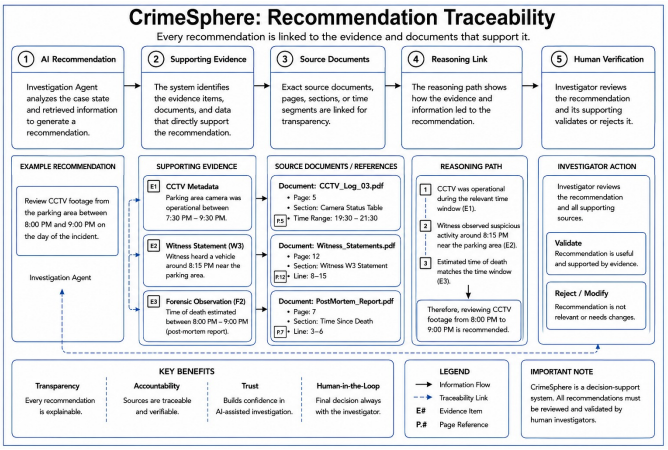


Fig. 4. Traceability of an AI-generated recommendation to supporting information.

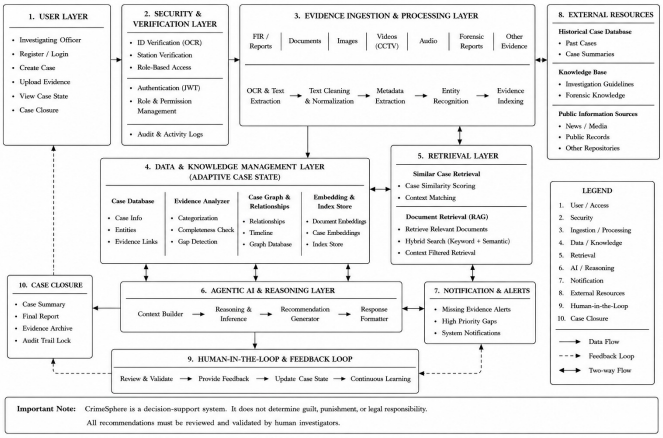


Fig. 5. Architecture of the proposed CrimeSphere framework.

IV. Retrieval and Agentic Reasoning

A. Retrieval Layer

Retrieval operates against the current case state along two directions. The first is similar-case retrieval, using semantic embeddings [12] and dense vector search [11] to find historically similar cases based on description, crime type, weapon, location, evidence characteristics, and modus operandi. The second is retrieval of current, publicly available information relevant to an ongoing case, obtained through a news or search API and kept visually and structurally separate from verified case evidence.

B. Retrieval-Augmented Generation and Investigation Agent

The context supplied to the investigation agent is built in stages: the current case state is serialized into a structured summary, this summary drives the retrieval layer described above, and the retrieved documents, historical cases, and any externally retrieved information are combined with the case-state summary, following the general RAG approach of Lewis et al. [10]. Because the case state changes as evidence is added, this context is reconstructed at each update rather than fixed once at the start, and the ordering of retrieved items follows the position sensitivity reported by Liu et al. [14].

The investigation agent, implemented using a large language model accessed through the Groq API, operates on this retrieval-augmented context to assist with case summarization, evidence organization, identification of potentially missing information, interpretation of similar cases, identification of relationships between case entities, and suggestions for investigative actions. This role is consistent with the broader literature on using language models as reasoning components that select and act on relevant information [15], [16], rather than as static text generators; CrimeSphere does not expose or depend on any hidden chain-of-thought process [17]. Its output is framed as assistance rather than a conclusion about the case; the investigator is expected to weigh each suggestion against their own knowledge of the case before acting on it.

TABLE II
Comparison of Existing Investigation Approaches and CrimeSphere

Approach	AI Investigation	RAG	Similar Case Retrieval	Adaptive Case State	Evidence Completeness	Agentic Reasoning	Traceability
Ding [1]	✓	—	✓	—	—	—	—
Skipanes et al. [2]	✓	✓	—	—	—	—	—
LAPIS [3]	✓	—	—	—	—	Limited	—
Bellandi et al. [4]	✓	—	✓	—	—	—	✓
Wang et al. [5]	✓	—	—	—	—	Limited	—
CrimeSphere	✓	✓	✓	✓	✓	✓	✓

V. Implementation

The investigator-facing interface is implemented in React with TypeScript, covering officer verification, FIR and case management, evidence upload, and the case dashboard. The backend is implemented in Spring Boot and exposes authentication, case and FIR management, evidence handling, OCR processing, and report generation through a REST API, as shown in Fig. 5.

Structured case information, including police officer and station records, FIR and case data, victim and suspect details, evidence metadata, and investigation records, is stored in PostgreSQL. Photographs, CCTV footage, forensic reports, post-mortem reports, and other documents are stored using file or object storage, with metadata linking each item to its case. Current news and public information related to an ongoing case are retrieved through a news or search API and displayed with source links; such material is never treated as verified case evidence.

When an investigation is completed, CrimeSphere organizes the FIR information, victim and suspect information, evidence, weapon information, investigation findings, case outcome, photographs, and supporting documents into a structured final closing report, generated as a PDF, as outlined in Fig. 6.

Table II summarizes how CrimeSphere compares against the approaches discussed in Section II, based on capabilities reported in the corresponding papers.

VI. Experimental Methodology

CrimeSphere's adaptive case-state approach was evaluated against a baseline performing static, document-based RAG

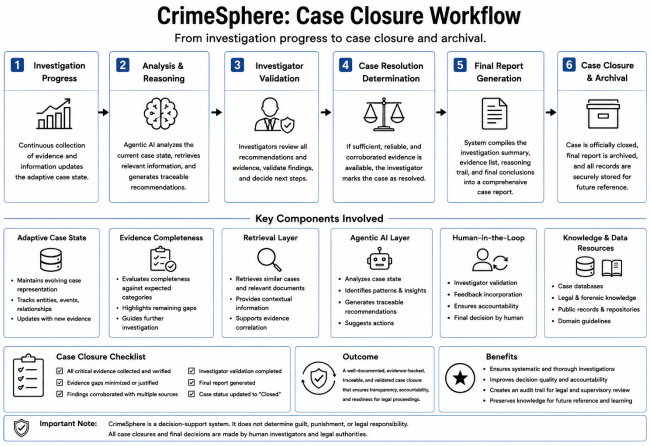


Fig. 6. Case closure and final investigation report generation.

over the same case data, holding retrieval and language-model components constant so any difference reflects the adaptive case-state mechanism rather than the underlying model. The evaluation uses a controlled set of 20 fictional murder-investigation scenarios (C01–C20), not derived from real cases or police records. Each scenario defines the expected information categories, categories intentionally withheld at evaluation time, a pool of candidate documents, and the recommendations a fully informed investigator would consider reasonable.

For each scenario, the same predefined case information is supplied to both Static RAG and CrimeSphere, and each system's output, consisting of the evidence gaps it flags, the documents it retrieves, the recommendations it generates, and any supporting evidence attached to those recommendations, is compared against the scenario's predefined expected gaps, expected relevant documents, and expected recommendations. This yields true positives, false positives, and false negatives for evidence-gap detection, counts of relevant versus retrieved items, a check of whether each recommendation's attached evidence actually supports it, and a relevance rating on a 1–5 scale. These per-scenario observations are aggregated across all 20 scenarios to compute the metrics in Table IV. Fig. 7 summarizes the controlled evaluation procedure used to compare the two approaches.

For each scenario, the expected evidence gaps, retrieved information, recommendations, and supporting evidence links are recorded and compared with the corresponding system outputs before the aggregate metrics are calculated. Because of the page limit of this conference paper, the full case-level records are retained as supporting evaluation material rather than reproduced here.

Evidence Gap F1 is computed from true/false positives and negatives between flagged and genuinely withheld categories:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{1}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

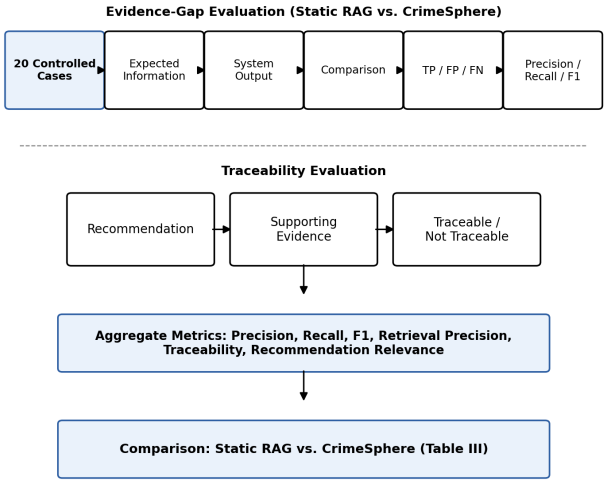


Fig. 7. Controlled evaluation procedure used to compare Static RAG and CrimeSphere.

TABLE III
INTERPRETATION OF EVALUATION METRICS

Metric	Interpretation
Evidence Gap F1	Balance between correctly identified and missed evidence gaps.
Retrieval Precision	Fraction of retrieved information that is relevant to the current case.
Traceability	Fraction of recommendations that can be linked to supporting evidence.
Recommendation Relevance	Evaluator rating of the usefulness and relevance of recommendations on a 1–5 scale.

Retrieval Precision is the number of relevant retrieved items over the total number of retrieved items. Traceability is the number of traceable recommendations over the total number of recommendations. Recommendation Relevance is the sum of relevance scores over the number of ratings, reported on a 1–5 scale [13].

Table III gives a short description of what each metric measures; the pilot numbers obtained by running this procedure are reported and discussed in Section VII.

VII. RESULTS AND DISCUSSION

Table IV lists the pilot numbers produced by the procedure described in Section VI. Both systems were given the same case information for each of the 20 scenarios, and what each one flagged as an evidence gap, what it retrieved, and what it recommended was checked against the scenario's predefined expectations: gap flags were scored with precision, recall, and F1; retrieved items were checked for relevance; each recommendation was checked for a supporting evidence link; and relevance was rated on a 1–5 scale. These per-scenario

TABLE IV
EXPERIMENTAL EVALUATION RESULTS

Method	Gap F1	Ret. Prec.	Trace.	Rec. Rel.
Static RAG	0.632	0.68	0.54	3.4 / 5
CrimeSphere	0.879	0.84	0.92	4.2 / 5

counts were then combined across all 20 scenarios to obtain the numbers below.

CrimeSphere scores higher than Static RAG on every metric: 0.879 against 0.632 for Evidence Gap F1, 0.84 against 0.68 for Retrieval Precision, 0.92 against 0.54 for Traceability, and 4.2 against 3.4 out of 5 for Recommendation Relevance. None of the four components, the adaptive case state, the completeness check, contextual retrieval, the agent, or the evidence links attached to recommendations, was tested on its own, so the gap between the two systems reflects how they work together rather than the contribution of any single part.

The Evidence Gap F1 difference comes from how the two systems treat new information. Static RAG checks the case against a document set that stays fixed once retrieval has run. CrimeSphere instead re-checks the case state each time it is updated, so a category flagged as missing earlier can be marked available once the corresponding evidence is entered.

Retrieval Precision is higher for CrimeSphere because retrieval is re-run against the current case state rather than a static index built once at the start. If the case has moved on, for example, a suspect has been ruled out, the retrieval context reflects that change instead of continuing to pull documents relevant to an earlier version of the case.

Traceability shows the largest gap between the two systems (0.92 versus 0.54), and it is the metric that says the most about what CrimeSphere is built to do. Every recommendation the investigation agent produces is stored together with the specific evidence item, document, or record it was generated from, so an investigator can open a recommendation and check what it is based on rather than take it on trust. Static RAG assembles a context window for each query, but nothing ties a generated recommendation back to a specific document once that context is discarded, which is why its traceability score is lower.

Recommendation Relevance is rated by the evaluator on a 1–5 scale, and CrimeSphere's average of 4.2 is higher than Static RAG's 3.4 in this pilot run. This score reflects one evaluator's judgment of how useful a recommendation looked, not a check of whether it was factually correct, so it needs to be read together with the other three metrics rather than on its own.

Fig. 8 shows a single run of CrimeSphere: the case state at that point, the evidence gaps it flagged, what it retrieved, the recommendations it generated, and the evidence linked to each one. This is one example meant to show what the system's output looks like in practice. It is separate from the aggregate numbers in Table IV and is not, by itself, evidence for those numbers.

The system can still get things wrong. Case information that is incomplete or ambiguous can lead to a missed or a false evidence-gap flag, a short or vague description can

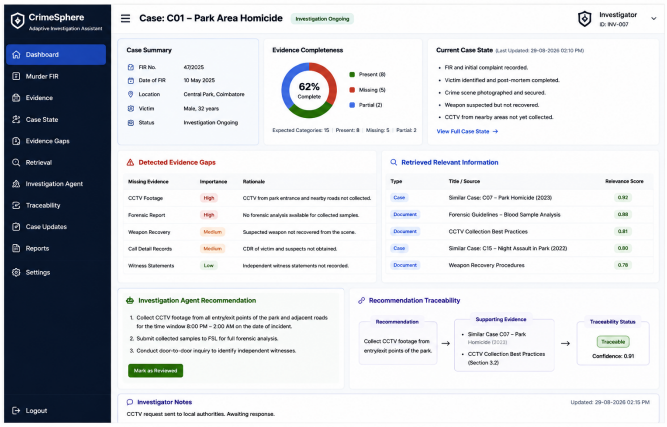


Fig. X. Representative CrimeSphere output showing evidence-gap identification, retrieved information, and evidence-linked recommendations.

Fig. 8. Representative CrimeSphere output showing evidence-gap identification, retrieved information, and evidence-linked recommendations.

pull in retrieved documents that are not actually relevant, and because the investigation agent is an LLM, it can produce a recommendation that reads plausibly but is not backed by anything, which is why every output still needs to be checked by the investigator. The numbers reported here come from a pilot run and are meant to show how the evaluation procedure works and how the two systems compare directionally. The full 20-scenario run has not been completed end-to-end, no significance testing has been done on these numbers, and they should not be read as a measure of how CrimeSphere would perform in an actual investigation.

VIII. FUTURE WORK

The next step is to extend the case state beyond text to handle images, CCTV or video footage, and documents together, instead of treating each type of evidence separately. A structured case graph is planned to capture how entities in a case, victims, suspects, evidence items, locations, relate to each other, which the current category-based case state does not represent directly. The 20-scenario pilot needs to be run to completion and extended with a larger, more varied set of cases reviewed by practicing investigators rather than a single evaluator. Beyond that, evidence provenance needs tighter tracking, and RAG-specific evaluation methods [13] should be applied as the retrieval component is tested further.

IX. CONCLUSION

Murder investigations involve information that keeps changing as FIRs, victim and suspect records, photographs, CCTV footage, and forensic reports are added over time. CrimeSphere addresses this by maintaining an adaptive case state that updates whenever new evidence arrives, checking the state for evidence completeness, and using retrieval and retrieval-augmented generation with an investigation agent to produce recommendations linked to their supporting evidence. Every recommendation is reviewed and validated by the investigator before any action is taken, and CrimeSphere does not determine guilt, punishment, or legal responsibility. Future work

includes evaluating the framework on larger, more diverse case datasets with domain-expert investigators.

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